Assignment No 8

Aim : Apply regularization to avoid overfitting.

Theory:

Regularization is a machine learning technique used to **reduce overfitting** and improve the generalization ability of a model. Overfitting occurs when a machine learning model learns the training data too closely, including its noise and unnecessary patterns, and therefore performs well on training data but poorly on unseen or test data.

Regularization works by adding a **penalty term** to the model's loss or cost function. This penalty discourages the model from assigning excessively large weights to features and helps produce a simpler model.

Overfitting

A model is said to be **overfitted** when it has very high performance on training data but significantly lower performance on testing or new data.

For example:

- Training accuracy = 98%
- Testing accuracy = 75%

This indicates that the model may have learned patterns specific to the training dataset instead of learning general patterns.

Regularization helps control this problem.

Types of Regularization

The two most commonly used regularization techniques are:

1. L1 Regularization — Lasso

L1 regularization adds the absolute values of the model coefficients to the loss function.

$$Loss = Original\ Loss + \lambda \sum |w_i|$$

Where:

- $\mathbf{w_i}$ = model coefficients or weights
- **λ (lambda)** = regularization parameter

L1 regularization can force some coefficients to become **zero**. Therefore, it can also perform **feature selection** by removing less important features.

Lasso Regression is an example of L1 regularization.

2. L2 Regularization — Ridge

L2 regularization adds the squared values of the model coefficients to the loss function.

$$Loss = Original\ Loss + \lambda \sum w_i^2$$

L2 regularization reduces the magnitude of the coefficients without generally making them exactly zero.

Ridge Regression is an example of L2 regularization.

Comparison of L1 and L2

L1 Regularization	**L2 Regularization**
Also called Lasso	Also called Ridge
Uses absolute values of coefficients	Uses squared values of coefficients
Can make coefficients exactly zero	Usually keeps coefficients non-zero
Performs feature selection	Helps control large coefficients
Useful when some features may be irrelevant	Useful when many features contribute to the prediction

Elastic Net

Elastic Net combines both L1 and L2 regularization. It is useful when a dataset contains many features and some of them may be correlated.

It can be represented as a combination of:

$$L1 + L2$$

This allows Elastic Net to perform feature selection while also controlling the magnitude of coefficients.

Role of Lambda (λ)

The **regularization parameter λ** controls the strength of the penalty.

- **Small λ:** Less regularization and the model can become more complex.
- **Large λ:** Stronger regularization and simpler model.
- **Very large λ:** May cause **underfitting**.

Therefore, an appropriate value of λ should be selected, commonly using techniques such as **cross-validation**.

How Regularization Reduces Overfitting

Without regularization, a model may assign very large weights to certain features in order to fit the training data closely.

Regularization adds a penalty for large weights:

Complex Model → Large Weights → Penalty → Smaller Weights → Simpler Model → Better Generalization

Thus, regularization helps the model focus on the most important patterns rather than noise in the training data.

Steps to Apply Regularization

1. Load the dataset.
2. Separate independent and dependent variables.
3. Split the dataset into training and testing sets.
4. Preprocess and scale the features if required.
5. Select a regularization technique such as L1 or L2.
6. Choose an appropriate regularization parameter.
7. Train the regularized model.
8. Make predictions on training and testing data.
9. Compare the model's performance.
10. Select the model that provides better generalization.

Advantages

- Reduces overfitting.
- Improves model generalization.
- Controls very large model coefficients.
- Can improve model stability.
- L1 regularization can perform feature selection.
- Useful for datasets containing many features.

Limitations

- Choosing the correct regularization strength can be difficult.
- Excessive regularization can cause underfitting.
- Regularization does not completely solve every cause of overfitting.
- Features may need to be scaled for some models.

Applications

Regularization is commonly used in:

- Linear Regression.
- Logistic Regression.
- Neural Networks.
- Machine learning models with many features.
- High-dimensional datasets.
- Prediction and classification problems.

Conclusion

Regularization is an important technique for preventing overfitting in machine learning. It adds a penalty to the model's loss function to control large coefficients and encourage simpler models. **L1 (Lasso), L2 (Ridge), and Elastic Net** are commonly used regularization techniques that improve the model's ability to perform well on unseen data.

Code:

```
import numpy as np

import matplotlib.pyplot as plt

from sklearn.svm import SVR

from sklearn.preprocessing import StandardScaler

X = np.array([[1], [2], [3], [4], [5], [6], [7], [8], [9], [10]])

y = np.array([1.2, 1.9, 3.2, 3.8, 5.1, 5.8, 7.2, 7.9, 9.1, 10.2])


sc_X = StandardScaler()

sc_y = StandardScaler()
```

```
X_scaled = sc_X.fit_transform(X)

y_scaled = sc_y.fit_transform(y.reshape(-1, 1))

svr_regressor = SVR(kernel='rbf', C=10, epsilon=0.2, gamma=0.1)

svr_regressor.fit(X_scaled, y_scaled.ravel())

X_test = np.array([[6.5]])

X_test_scaled = sc_X.transform(X_test)

y_pred_scaled = svr_regressor.predict(X_test_scaled)

y_pred = sc_y.inverse_transform(y_pred_scaled.reshape(-1, 1))

print("Predicted value for X=6.5:", y_pred[0][0])

plt.scatter(X, y, color='red', label='Original Data')

plt.plot(X, sc_y.inverse_transform(svr_regressor.predict(X_scaled).reshape(-1, 1)),

        color='blue', label='SVR Prediction (Regularized)')

plt.title('SVR Regression with Regularization')

plt.xlabel('X')

plt.ylabel('y')

plt.legend()

plt.show()
```

Output:

